Persistent Homology of the SYK / JT Correspondence: A Structure Theorem and a Negative Result at Accessible N

The barcode comparison reduces exactly to two correlator scalars per side.
At every system size we can reach, the declared large-N dictionary is falsified,
and the mechanism is that no accessible cell is semiclassical.

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Working draft. All figures and numbers are taken from the committed results files (`rung1_results.txt`–`rung4_results.txt`, `rung4_checks_results.txt`).

Abstract. We ask whether the Sachdev–Ye–Kitaev (SYK) model and its Jackiw–Teitelboim (JT) gravity dual produce isomorphic persistence modules when their thermal two-point structure is compared through Vietoris–Rips persistent homology, a topological-data-analysis probe of the ER=EPR correspondence. To our knowledge this is the first application of persistent homology to a holographic system with a known gravity dual comparing barcodes on both sides; the closest prior work [14] names this direction as open. We report three results. First, a structure result: for the natural thermal-circle construction the Vietoris–Rips persistence module is determined by two scalar correlator values per side, with no topological content beyond a single structurally guaranteed loop, so the statement "$H_1 > 0$ on both sides" cannot fail and carries zero information; this follows from Adamaszek–Adams applied to a strictly monotone circulant filtration and is verified identically across all 224 diagrams in our grid. Second, using a large-N literature value for the Schwarzian coupling C as a declared input, because it could not be measured from our own spectra, the SYK–JT comparison is a decisive negative at accessible sizes $N \leq 18$: the declared dictionary is falsified two independent ways, by temperature dependence of the fitted coupling (29–158% drift at fixed N) and by inconsistency between the two separations a single coupling must simultaneously match (ratios 1.47 to 7.81), with no large-N scaling ($b = 2.16 \pm 2.40$ against the predicted $b = 1$). Third, we locate the mechanism: at accessible N the SYK short-distance correlator is nearly free, the disagreement is ultraviolet-localized, and the entire accessible parameter range lies at $\beta/C \in [27.7, 332]$, deep in the strongly-quantum regime where a semiclassical JT dual is not expected to hold. The same conclusion is reached three independent ways. We frame these findings as the method's honest output: the construction is novel and correctly implemented, and what it detects at accessible N is that the system has not entered the regime in which the duality is sharp. We state precisely what is proven and what remains conjectural.

1 Introduction

1.1 The question

The ER=EPR proposal [12] identifies the entanglement structure of a quantum state with the geometry of a dual spacetime. The SYK model [16,10], N Majorana fermions with random all-to-all quartic couplings, and JT gravity [7,17] furnish the sharpest concrete arena: in the large-N, low-temperature limit SYK is governed by the same Schwarzian effective action as the boundary mode of JT gravity in nearly-AdS_2 [2,11,8].

Our question is topological. We ask whether the same persistent-homology fingerprint arises from the SYK correlation structure and from the JT geometric structure, and whether the map between the two barcodes is a genuine isomorphism of persistence modules or a coincidence of construction. Persistent homology [6,18,4] is the natural instrument: it is computable on both sides from a distance matrix alone, and the Wasserstein and bottleneck metrics give a principled notion of barcode proximity, with established stability properties [5].

1.2 The claim

> *A correspondence between entanglement and geometry, if it is exact at the level of topological structure, should present as an isomorphism of persistence modules. Whether a given construction can detect such an isomorphism is a separate question from whether the isomorphism*

exists, and the first must be settled before the second is asked. On the natural thermal-circle construction we settle the first, negatively: the construction cannot detect topology, because it has none to detect.

1.3 How this was established

The work is a ladder of five rungs of increasing sophistication; this draft covers the complete Rungs 1–4. A methodological commitment runs through all of them: *agreement between two computed objects is evidence only when the objects are genuinely independent*. Several apparent early successes were withdrawn once we established that the agreement was guaranteed by construction rather than discovered. That discipline is why the honest outcome at accessible N is a characterized negative rather than a claimed confirmation. Section 2 reports the withdrawals; they are part of the evidence.

1.4 Contributions

1. **A two-scalar structure result (§5).** For a strictly monotone circulant filtration on n points, the Vietoris–Rips persistence module is determined by $\widehat{G}(1)$ and $\widehat{G}(k^*)$ with $k^* = \lceil n/3 \rceil$. At $n = 24$: H_1 is exactly one bar $[1/\widehat{G}(1),\, 1/\widehat{G}(8)]$ and H_0 is exactly 23 finite bars $[0,\, 1/\widehat{G}(1)]$, for any monotone profile. Every Wasserstein number is then an exact function of two scalars per side, weighted $\approx 23{:}1$ toward $\widehat{G}(1)$. Verified across all 224 diagrams with zero violations. The methodological consequence is that on this construction persistent homology is not measuring topology.

2. **A falsification of the declared dictionary at accessible N (§6).** Fitting C per cell against 16 constraints with one parameter fails two independent ways: the fitted coupling drifts with temperature at fixed N by 29%, 158%, 93%, 35% where a genuine coupling would show $\approx 0\%$; and a single C cannot match both separations, with $C_{\text{fit}}(\widehat{G}_1)/C_{\text{fit}}(\widehat{G}_8) = 1.47, 7.81, 3.14, 1.86$ across N. The power-law fit gives $b = 2.16 \pm 2.40$ against the Maldacena–Stanford prediction $b = 1$.

3. **A mechanism, reached three independent ways (§6.3).** The declared dictionary forces $C \leq 0.18$ for $N \leq 18$, hence $\beta/C \in [27.7, 332]$ across the whole grid, so no accessible-N cell could have been semiclassical. Grid refinement makes agreement worse and excluding the shortest separations makes it better, so the disagreement is ultraviolet-localized where SYK is still nearly free. Independently, the coupling-extraction gate finds our spectra indistinguishable from a bandwidth-matched GOE control.

4. **A validated exact-Schwarzian implementation (§4.3).** Five independent checks, including a small-C validation at the $C \approx 0.1$–0.2 actually used, closing the gap between where the integrator was validated and where it is applied.

5. **A withdrawal record (§2).** Rungs 1–2 reproduce machine-precision agreement and are reported as framework validation only, with the tautology stated explicitly.

1.5 What this paper does not claim

Rung 4 says nothing about ER=EPR itself. It says that one construction, in one regime, does not detect the correspondence, and it identifies why. The categorical programme (Rung 5) is not addressed here.

2 Rungs 1–2: framework validation, not physics

Rung 1 compares the entanglement barcode of two maximally entangled qubits with the barcode of the dual wormhole throat. Rung 2 replaces the fixed state with a thermofield-double state and sweeps the inverse temperature β over 100 log-spaced values in $[0.01, 20.0]$. Numerically both agree to machine precision: the maximum Wasserstein distance over the Rung 2 sweep is 3.638×10^{-12} in exact double precision. The raw float32 `ripser` backend alone produces values up to 0.73; both are reported.

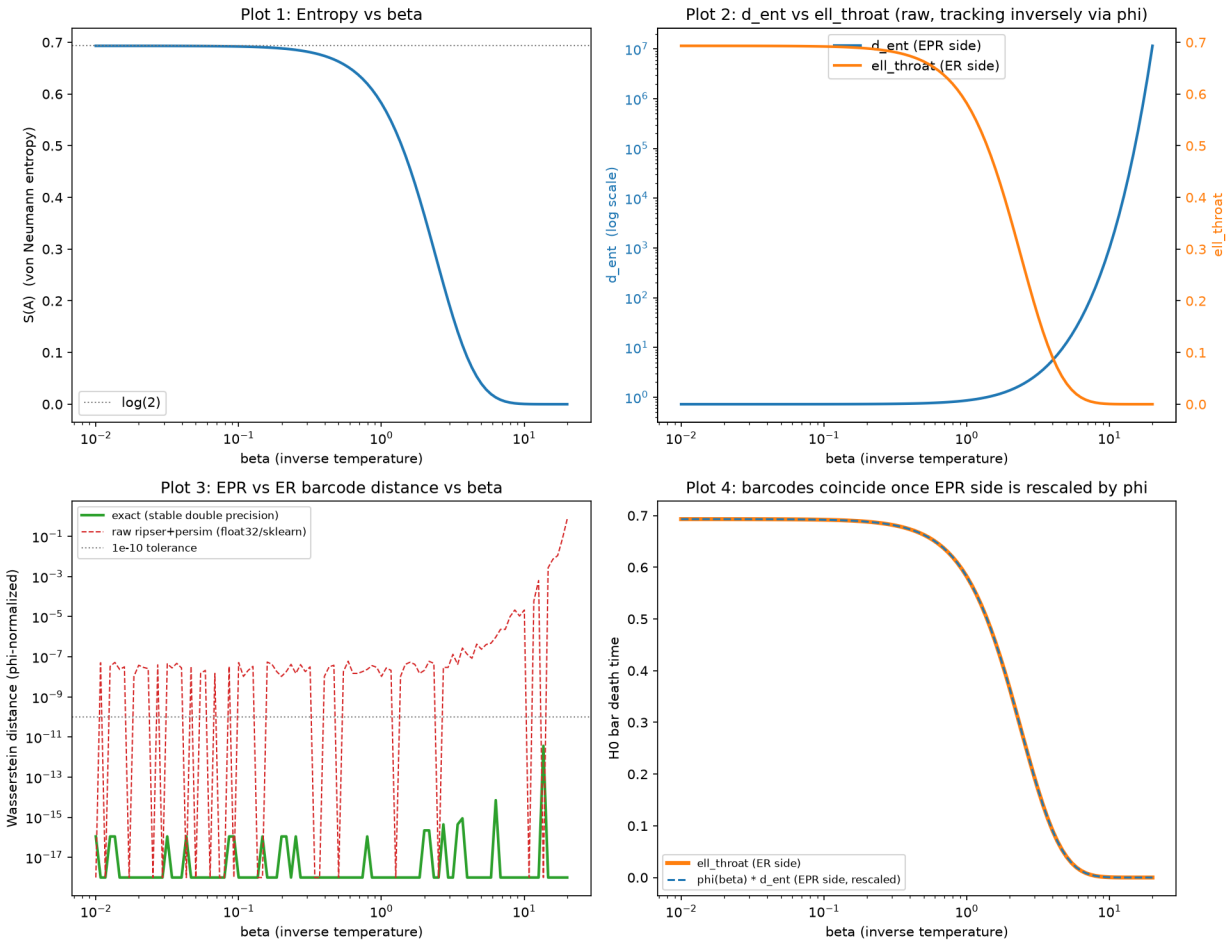


Figure 1: Rung 2 diagnostics. Entropy $S(A)$ vs β (a smooth analytic crossover, no transition); d_{ent} and ℓ_{throat} tracking inversely; the barcode distance at machine precision in exact double precision (green) while the raw float32 pipeline (red) spikes above tolerance; and the EPR barcode landing exactly on the ER barcode once rescaled by φ. The coincidence in the lower-right panel is the tautology visualized: it is guaranteed, not discovered.

This is not a physics result, for two independent reasons.

The agreement is tautological. The map φ was defined as the ratio of two quantities defined to be equal: the JT throat length was set to $S(A)$ and the entanglement distance to $1/I(A{:}B)$. With φ their ratio, Wasserstein zero is algebraically inevitable at every temperature. The results file makes this explicit: the identity $\varphi = S(A) \cdot I(A{:}B) = 2S(A)^2$ holds with maximum deviation 0.0 across the entire sweep. Ryu–Takayanagi [15] entered as an input, not an output.

The distance function is not a metric. The construction used $d(i,j) = 1/|C_{ij}|$, giving $d(i,i) = 2 \neq 0$. Any H_1 from these rungs is therefore suspect on structural grounds independent of the tautology. With two points the Betti numbers also never change under the sweep: $S(A)(\beta)$ is a smooth analytic sigmoid, a crossover set by the gap $E_1 - E_0 = 1$, not a topological transition.

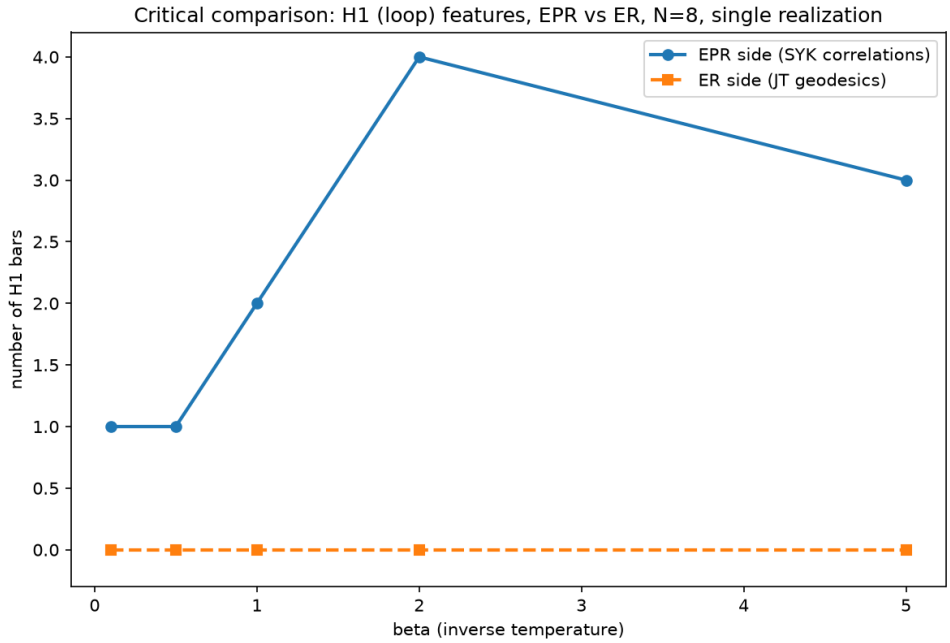


Figure 2: The Rung 3 obstruction. The EPR side produces a temperature-dependent number of H_1 loops (rising to 4 near $\beta = 2$), while the ER side is pinned at exactly zero for every β — provably, by the chordal-graph argument. The comparison is structurally lopsided.

3 Rung 3: the indexing obstruction and an honest null

Rung 3 moves to the full SYK model ($N = 8$ Majorana fermions via Jordan–Wigner, algebra verified to 2.2×10^{-16}; random four-body couplings with variance $6J^2/N^3$). The single-sided $\langle \chi_i^L \chi_j^L \rangle$ correlator was found exactly degenerate (diagonal mass 2.6×10^{-15}), so the EPR side uses the two-sided $C_{ij} = \langle \mathrm{TFD} | \chi_i^L \chi_j^R | \mathrm{TFD} \rangle$ with $d = 1/|C_{ij}|$. The ER side, as originally constructed, was the geodesic distance matrix on the $t = 0$ spatial slice of the JT eternal black hole [9] (20 sample points).

This construction is ill-posed in two compounding ways.

1. **Index mismatch.** The index sets differ — EPR: fermion flavor; ER: radial position on the $t = 0$ slice — and are not related by the holographic dictionary. SYK is 0+1 dimensional; the flavor index has no bulk image, and N maps to $1/G_N$, a coupling, not a coordinate.

2. **Dimensional obstruction.** The $t = 0$ slice is one-dimensional. The Vietoris–Rips complex of points on a line is collapsible and has trivial H_1 at every scale, for any sample size. The correct proof of triviality is the chordal indifference-graph argument, not a curvature or triangle-filling argument.

3.1 The measured null

The EPR side carries genuine loop structure while the ER side is structurally barred from any. At $N = 8$, $\beta = 1$ the EPR side has $H_1 = 2$ finite bars and the ER side $H_1 = 0$ (with $W_{H_0} = 109.09$, $W_{H_1} = 23.59$, unnormalized). Across the sweep $\beta \in \{0.1, 0.5, 1, 2, 5\}$ the EPR H_1 counts are $[1, 1, 2, 4, 3]$ against ER $H_1 = 0$ at every β. The disorder average over 10 realizations gives EPR $H_1 = 2.80 \pm 1.17$ — explicitly not a robust signal — against ER $H_1 = 0.00 \pm 0.00$; an $N = 12$ check gives EPR $H_1 = 7$, ER $H_1 = 0$.

The verdict is structural, not a falsification of ER=EPR. It defines what a fair test requires: a τ-indexed construction in which both sides can carry the same topology. We stress the change of ER-side object between rungs: Rung 3's ER side is JT geodesics on a one-dimensional slice (hence $H_1 \equiv 0$); Rung 4's ER side is the

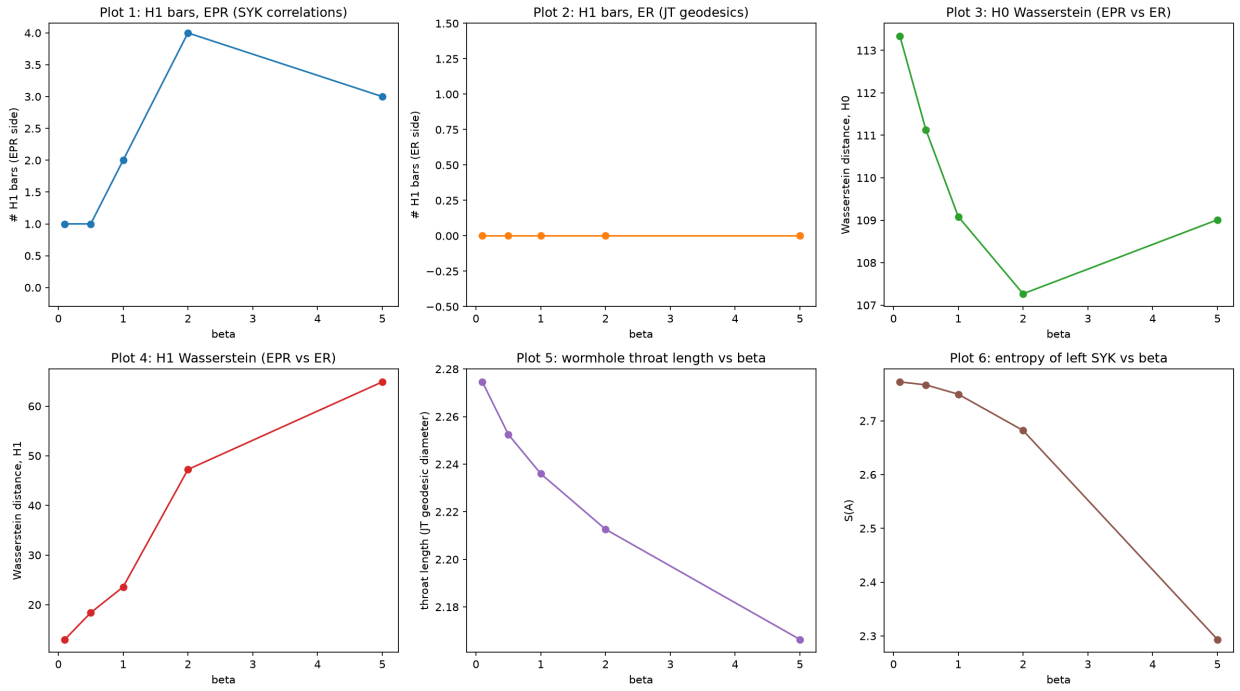


Figure 3: Rung 3 full diagnostics: EPR H_1 counts vs β, the flat ER $H_1 = 0$, the H_0 and H_1 Wasserstein distances (large and unnormalized, reflecting the structural mismatch rather than a physics signal), the JT throat length, and the SYK entropy.

exact Schwarzian correlator on the thermal circle, a genuinely different construction.

4 Rung 4: the τ-indexed comparison

4.1 The two sides

EPR (SYK). For each disorder realization we diagonalize H exactly and form the flavor-averaged thermal two-point function $G(\tau_a, \tau_b)$ at the distinct circle separations, from $\mathrm{Tr}[e^{-(\beta-\tau)H}\chi\, e^{-\tau H}\chi]/Z$ averaged over the N fermions. The construction respects fermion antiperiodicity and is positive on the strip; the standing identity $\sum_m W_{nm} = 1/2$ holds to $\sim 10^{-15}$.

ER (Schwarzian). We use the exact finite-temperature Schwarzian two-point function (Mertens–Turiaci–Verlinde [13]), with spectral density $\rho(k) = k\sinh(2\pi k)$, the four-Gamma matrix element

$$\frac{|\Gamma(\Delta + i(k_1+k_2))|^2\,|\Gamma(\Delta + i(k_1-k_2))|^2}{\Gamma(2\Delta)}, \qquad \Delta = \tfrac{1}{4},$$

and thermal weights $e^{-\tau k_1^2/2C-(\beta-\tau)k_2^2/2C}$, computed by a validated two-dimensional quadrature in $(s,d) = (k_1+k_2, k_1-k_2)$ coordinates with change-of-variables Jacobian $-\log 2$. This is A/B verified: toggling shifts $\log G$ by exactly $\log 2$ to 5.5×10^{-14}, leaving the shape-gauged barcode bit-for-bit identical.

4.2 The distance, the gauge, and the grid

Both sides are placed on the same 24-point thermal circle. Distances use $d = 1/\widehat{G}$ with $\widehat{G} = G/G(\beta/2)$, the *shape gauge*, so both sides equal 1 at maximal separation and the comparison probes correlation-decay shape. Persistent homology is computed with `ripser` [3] at maxdim $= 1$; H_0 and H_1 Wasserstein distances

are computed per realization ($R = 10$, seeds $1000N + r$) against declared-C, mismatched-C ($\times 5$, $\div 5$), and conformal-limit references, over $N \in \{12, 14, 16, 18\} \times \beta \in \{5, 10, 20, 40\}$.

4.3 ER-side validation

The Schwarzian correlator passed five independent checks within the validated scope $C \leq 200$ — a stated physics decision, not a silent workaround.

- **Check A, free-energy calibration.** $\ln Z$ matches the Schwarzian asymptotic with the true ρ settling to a constant (drift 1.7×10^{-13}) while the wrong-power controls (k^0, k^2) drift by ≈ 3.46. PASS.
- **Check B, conformal reduction.** The correlator shape approaches the conformal limit at rate $O(\beta/C)$: per-C-doubling ratio 0.500, C·deviation constant to four digits, log-log slope 1.000. PASS.
- **Check C, barcode collapse.** The Schwarzian barcode collapses monotonically onto the tree-level frozen-circle barcode as C grows (Wasserstein halving $1.64 \times 10^{-4} \to 8.2 \times 10^{-5} \to 4.1 \times 10^{-5} \to 2.05 \times 10^{-5}$). PASS.
- **Check D, β/C is the real knob.** G depends only on β/C (0.00 spread at fixed β/C across different (β, C)) and genuinely varies when β/C changes. PASS.
- **Check E, small-C validation.** Because the comparison lives at $C \approx 0.1$–0.2, far below the $C \geq 25$ of Checks A–D, an independent dense-grid cross-check confirms `compute_G` there to $\sim 10^{-16}$ relative agreement. PASS. This closes the regime-mismatch gap between where the integrator was validated and where it is used.

4.4 Why C is declared, not measured

The Schwarzian coupling C is the ER side's only free parameter. We attempted to measure it from the SYK spectra via a pre-registered gate with two independent routes and a random-matrix (GOE) control.

- **Route 1**, density-of-states sinh-edge fit: the slope never plateaued across quantile bands at any N. No C exists to extract.
- **Route 2**, entropy linear-in-T fit: fitted C was negative at every N.
- The two routes disagreed by 230–433%, and — decisively — SYK's fit pattern tracked the bandwidth-matched GOE control almost exactly. The accessible spectral edge is generic random-matrix behavior; the Schwarzian coupling has not emerged at these sizes.

The gate failed $0/4$. We therefore declare C from Maldacena–Stanford [10]: $C(N) = \sqrt{2}\,\alpha_S N$ with $\alpha_S = 0.00709$, derived from the stated $q = 4$ specific-heat coefficient $c \approx 0.396 N/\mathscr{J}$ via $c = 4\pi^2 \alpha_S N/\mathscr{J}$ and $\mathscr{J} = J/\sqrt{2}$. We note that α_S is a single-source constant: all published $q = 4$ values trace to this fit. Rung 4 is therefore an application of the large-N dictionary at small N — an imported calibration, not a measurement.

5 Structure: the comparison reduces to two scalars

Definition 1 (Monotone circulant filtration)**.** *Let $\widehat{G} : \{0, 1, \ldots, n-1\} \to \mathbb{R}_{>0}$ be a function of circle separation m alone, with $\widehat{G}(0) = 1$ and $\widehat{G}$ strictly decreasing in m on $0 \leq m \leq n/2$ and symmetric about $n/2$. The induced distance matrix $D_{ab} = 1/\widehat{G}(|a-b| \bmod n)$ is circulant, and the Vietoris–Rips filtration of D is the* monotone circulant filtration *on n points.*

Both sides of the Rung 4 comparison are of this form. The matrix depends only on circle separation by construction, and $\widehat{G}$ is strictly monotone in m on both sides — verified across all 224 diagrams in the grid with zero violations, and guaranteed analytically by the spectral representation.

Proposition 1 (Two-scalar reduction)**.** *For a monotone circulant filtration on n points, the Vietoris–Rips filtration passes through exactly the nested clique complexes of the circulant graphs, and the persistence module is determined by the two values $\widehat{G}(1)$ and $\widehat{G}(k^*)$ with $k^* = \lceil n/3 \rceil$. Concretely, at $n = 24$:*

- *H_1 is exactly one bar on each side, $[\,1/\widehat{G}(1),\ 1/\widehat{G}(8)\,]$, for any monotone profile — never empty, never a second bar;*
- *H_0 is exactly $n-1 = 23$ finite bars, all $[\,0,\ 1/\widehat{G}(1)\,]$.*

Consequently every Wasserstein number is an exact function of $(\widehat{G}(1), \widehat{G}(8))$ per side, weighted $\approx 23{:}1$ toward $\widehat{G}(1)$.

Proof sketch. Strict monotonicity means the sublevel sets of D are exactly the circulant graphs C_n^k for increasing k, so the Vietoris–Rips filtration is the nested family of their clique complexes, indexed by k rather than by a continuum. The homotopy types of these complexes are given by the Adamaszek–Adams classification [1] of Vietoris–Rips complexes of the circle: the complex is homotopy equivalent to an odd sphere or a wedge of even spheres, with S^1 occurring precisely for $1 \le k < n/3$. Taking H_1 therefore gives a single bar, born when the nearest-neighbor edge enters at $1/\widehat{G}(1)$ and dying when k reaches $k^* = \lceil n/3 \rceil$, at $1/\widehat{G}(k^*)$. H_0 is immediate: all n vertices merge simultaneously when the nearest-neighbor scale is crossed. □

Status. Proposition 1 is stated at the level of the construction together with grid-wide numerical verification: the predicted single H_1 bar and 23 H_0 bars are reproduced on all 224 diagrams to $\approx 3 \times 10^{-8}$, and the weighting identity $W_{H_0} = 23 \cdot |\Delta(1/\widehat{G}(1))|$ holds to 7.7×10^{-7} (persim $p = 1$, L^2 ground metric). A formal statement and proof to publication standard are in preparation and are not claimed here.

Corollary 1 (The comparison is not topological)**.** *On this construction, persistent homology is not measuring topology. The statement "$H_1 > 0$ on both sides" cannot fail and carries zero information; the comparison is a two-scalar correlator comparison in topological clothing.*

This is a deeper instance of the discipline that withdrew Rungs 1–2, and we regard it as the paper's principal methodological contribution: any attempt to compare holographic duals through Vietoris–Rips persistence on a single monotone thermal circle will reduce the same way.

6 Results: the declared dictionary is falsified at accessible N

6.1 Reachability is per-cell and temperature-dependent

$\widehat{G}_{\text{Schwarzian}}(1)$ is a function of β/C alone, monotone increasing, with infimum equal to the conformal value 2.7679. SYK's $\widehat{G}(1)$ grows with β — for $N = 18$: 1.303, 1.729, 2.474, 3.554 at $\beta = 5, 10, 20, 40$ — and crosses the conformal value between $\beta = 20$ and $\beta = 40$ (crossing $\beta \approx 21$–27 across N). Hence for all 12 cells at $\beta \le 20$, SYK's $\widehat{G}(1)$ lies below the conformal infimum, unreachable by any C, and only the 4 cells at $\beta = 40$ admit a fit at all.

6.2 Where a fit exists, it is falsified two independent ways

C is a coupling, $C = \sqrt{2}\,\alpha_S N$, and cannot depend on temperature. Fitting C per cell independently — 16 constraints, one parameter — is the falsifiable test.

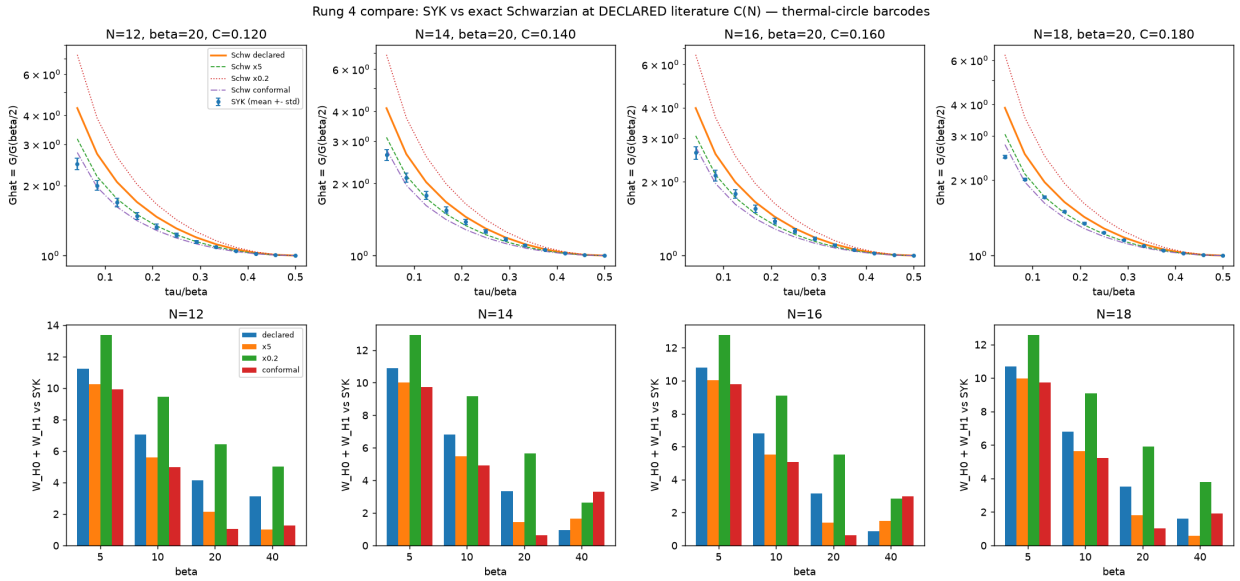


Figure 4: Rung 4 comparison, $N \in \{12, 14, 16, 18\}$. Top row: shape-gauged $\widehat{G}(\tau)$ vs τ/β. The SYK data (blue points) sit systematically below even the conformal control (purple), while the declared-C Schwarzian (orange) is steeper still — SYK is flatter than conformal at short distance. Bottom row: total $H_0 + H_1$ Wasserstein vs SYK for declared (blue), $\times 5$ (orange), $\div 5$ (green), conformal (red).

1. **Temperature dependence.** On the $\widehat{G}(8)$ target, where fits exist at two temperatures, C_{fit} drifts with β at fixed N by 29% ($N = 12$), 158% ($N = 14$), 93% ($N = 16$), 35% ($N = 18$). A genuine coupling would show $\approx 0\%$.

2. **Two-scalar inconsistency.** One C cannot match both separations: $C_{\text{fit}}(\widehat{G}_1)/C_{\text{fit}}(\widehat{G}_8) = 1.47, 7.81, 3.14, 1.86$ across N, systematically off unity. One profile cannot be fit by one parameter at two separations, so the Schwarzian *shape* is wrong, not merely miscalibrated.

3. **No large-N scaling.** The power-law fit $C_{\text{fit}} = aN^b$ on the $\beta = 40$ cells gives $b = 2.16 \pm 2.40$ — uncertainty exceeding the value, non-monotone in N, $\chi^2/\text{dof} = 34.8/2$ — against the Maldacena–Stanford prediction $b = 1$, $a = 0.01003$. There is no coupling to extract.

6.3 Mechanism: an ultraviolet, strongly-quantum story

These findings cohere. The β/C values span 27.7 to 332 across the grid — one and a half to two and a half orders of magnitude on the strongly-quantum side of $\beta/C = 1$, where the $O(\beta/C)$ semiclassical expansion is valid — and this is forced by the declared dictionary itself ($C \leq 0.18$ for $N \leq 18$), so no accessible-N cell could have been semiclassical. Grid refinement makes agreement worse (per-pair gap $0.340 \to 0.445 \to 0.523$ as $n_\tau = 12 \to 24 \to 48$) and excluding the shortest separations makes it better (W_{H_0}: $10.24 \to 7.82 \to 6.05$): the disagreement is ultraviolet-localized, in the nearest-neighbor correlation where SYK is still nearly free, and the metric is weighted 23:1 toward exactly that scale.

This is the same conclusion reached three independent ways: the extraction gate (RMT-like spectra), the β/C accounting (whole grid strongly-quantum), and the correlator itself.

7 What is proven and what is conjectured

Established here.

- SYK $G(\tau)$ machinery correct: $G(0) = 1/2$ and flavor-weight row sums $1/2$ to machine precision; $G > 0$

on the strip.

- Schwarzian correlator correctly implemented and regularized: Checks A–E all pass, including the small-C validation at the $C \approx 0.1$–0.2 used in the comparison; Jacobian A/B-verified.
- On the monotone circulant construction the persistence module is two scalars per side, so H_1-existence carries no information — by construction plus grid-wide verification (Proposition 1; formal proof in preparation).
- The C-extraction gate fails $0/4$: SYK spectra are GOE-like at $N \leq 18$.
- Where a fit exists, the declared dictionary is falsified by temperature dependence and by two-scalar inconsistency, with no large-N scaling.

Not established here.

- Whether the duality becomes sharp — barcodes agreeing non-tautologically — at N well beyond 18. The literature suggests Schwarzian-edge onset near $N \approx 24$–34; $N = 20$–24 is proposed future work.
- Whether persistent homology of a *single-boundary* thermal-circle distance matrix is the right invariant at all, given the two-scalar reduction. A two-boundary construction, in which the connectivity of the Einstein–Rosen bridge enters the filtered object rather than being traced away, is the natural next candidate.
- Any statement about the categorical natural-transformation programme (Rung 5), which is not addressed in this draft. **Rung 4 says nothing about ER=EPR itself.**

8 Rung 5: the categorical statement, not completed

The programme this paper belongs to has five rungs. Rungs 1–4 are complete and are what we report above. Rung 5 is not completed, and nothing in this draft should be read as progress on it. We state here what it requires, and what Rungs 1–4 have changed about what it would need to look like.

8.1 What Rung 5 requires

The intended statement is categorical. Write $\mathscr{P}_{\mathrm{EPR}}$ for the persistence module obtained from the SYK side and $\mathscr{P}_{\mathrm{ER}}$ for the one obtained from the gravity side, both as functors

$$\mathscr{P} : (\mathbb{R}_{\geq 0}, \leq) \longrightarrow \mathbf{Vect}_k, \qquad \varepsilon \longmapsto H_\bullet(\mathrm{VR}_\varepsilon(\,\cdot\,)),$$

with the internal maps induced by inclusion of filtration steps. Rung 5 asks for a natural transformation $\varphi : \mathscr{P}_{\mathrm{EPR}} \Rightarrow \mathscr{P}_{\mathrm{ER}}$ and a proof that it is a natural *isomorphism*. Three things are required:

1. **Construction of φ.** An explicit family of maps $\varphi_\varepsilon : \mathscr{P}_{\mathrm{EPR}}(\varepsilon) \to \mathscr{P}_{\mathrm{ER}}(\varepsilon)$, derived from the holographic dictionary rather than fitted.
2. **Functoriality.** That the construction respects the filtration order, so that φ is a morphism of persistence modules rather than a scale-by-scale coincidence.
3. **Invertibility.** That each φ_ε is an isomorphism, which by the structure theorem for persistence modules over a field is equivalent to the two barcodes being equal as multisets of intervals.

We note that naturality is not an additional burden in this setting: morphisms of persistence modules are by definition natural transformations between the corresponding functors, so no separate verification is required. The content of Rung 5 lies entirely in constructing φ and establishing invertibility.

8.2 Why Rungs 1–4 change what Rung 5 must be

Proposition 1 constrains the programme sharply. On the monotone circulant construction used in Rung 4, $\mathscr{P}_{\mathrm{EPR}}$ and $\mathscr{P}_{\mathrm{ER}}$ each decompose into exactly one H_1 interval and $n-1$ identical H_0 intervals, all determined by $\widehat{G}(1)$ and $\widehat{G}(k^*)$. A natural isomorphism between them would therefore be the statement that two correlator scalars agree on each side — true or false as a matter of numerics, but carrying no topological content. **Rung 5, posed on this construction, would be vacuous even if it were proved.**

This is the sense in which the Rung 4 negative is informative rather than merely disappointing. It does not report that φ fails to exist; it reports that the category in which we were attempting to place the question is too coarse to hold it. A non-vacuous Rung 5 requires a filtered object with genuine topological content on both sides — the two-boundary construction of §7 is the natural candidate, since there the connectivity of the Einstein–Rosen bridge enters the filtration rather than being traced away — and it requires a regime in which the dual is sharp, which by §6.3 is not any $N \leq 18$.

8.3 Status

Rung 5 is open. We have not constructed φ, have not proved functoriality, and have not proved invertibility. The formal proof of Proposition 1, which would supply the structure-theorem step of any eventual invertibility argument, is likewise in preparation rather than complete. We record the target here so that the scope of the present claims is unambiguous: *this paper contains no categorical result*, and the ER=EPR conjecture itself is neither supported nor contradicted by anything above.

9 Conclusion

We built and validated a persistent-homology probe of the SYK/JT correspondence and report honestly what it detects at accessible sizes. The construction is correct; the physics it reveals at $N \leq 18$ is that the system is not yet in the regime where a semiclassical JT dual holds, so the declared large-N dictionary is falsified rather than confirmed — cleanly, two independent ways, with an ultraviolet mechanism. Along the way we established that the barcode comparison on this construction reduces exactly to two correlator scalars, itself a cautionary methodological result for topological data analysis in holography. A clean, mechanistically explained negative for a genuinely new method is the contribution; the strong version is not claimed.

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